1 Real CHIME data



981 bursts · 83 repeaters

Catalog 2, 400–800 MHz

→ 950 noise-free templates
→ 856 real noise backgrounds
sources split 60/20/20 by source

2 Pair simulator

lensed



same burst twice

repeat



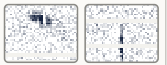
two bursts, one source

impostor



shared imprint

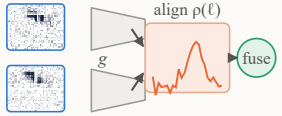
unrelated



two sources

+ nuisances per member: time shift, ΔDM, scattering, scintillation; frozen, hashed sets

3 Siamese correlation net



1. shared encoder, 2-sample resolution
2. soft alignment over ±24 samples
3. fuse $|g_A - g_B|$ and $g_A \odot g_B$
4. 4-class head → score s(A, B)

259k parameters

4 Validation, no real positives

- Real cross-source pairs vs per-pair twins — **FPR ≤ 1.5%**
- Real-burst copies vs template copies — **ratio 0.97**
- Second-machine rerun data, sets, training — **7/7 hashes**
- Grouped folds, sealed test pre-registered protocol — **OOF +0.22**

5 Deployment: where can the detector work?

Timing test on burst lists (false alarms checked on 166,000 scrambles) and survey-geometry simulation → copies are recorded for only ~1.8% of lenses at a transit telescope; an all-sky monitor watching the same sources would find ~50× more lensed repeaters.